

“Software optimisation helped us greatly reduce the chipset cost”



AllGo Embedded Systems was founded by professionals with a total of 70+ years of experience in DSP and embedded system development. Their solution development experience spans automotive infotainment systems, high-definition video, short-range wireless, consumer audio systems and CD players.

Darshak Vasavada, co-founder of AllGo Embedded Systems, speaks to Dilin Anand, technical correspondent at EFY, about Stamp—a low-cost customisable tablet developed by AllGo

DARSHAK VASAVADA
CO-FOUNDER, ALLGO EMBEDDED SYSTEMS

Please tell us about Stamp.

Stamp is a low-cost tablet designed for the Indian market. It is our first product fully developed from scratch including the hardware, circuits and mechanical design. Everything, except manufacturing, is done in-house.

Manufacturing was outsourced. But as we did not want to compromise on quality, all key components like the processor, LCD, battery, memory and power management circuits were identified by us.

How did you manage to keep the tablet's cost low?

We used a highly integrated processor based on ARM9 core. This processor—Freescale MX233—was originally intended for use in digital photo frames and Internet radios. We proceeded to port Android on top of this particular inexpensive processor.

We used our expertise in DSP and firmware to squeeze the performance out of MX233. For example, Android 2.2 requires a GPU, but our system-on-chip (SoC) does not have a GPU. So we optimised those parts of code which use software libraries for graphics.

Likewise, we used a pixel pipeline on the processor and some optimised video codecs to improve the performance. The processor itself was not designed to support video graphics array (VGA); the datasheet states that it can go up to VGA. We changed it so that it can support wide VGA.

In short, we picked up an integrated device and extracted better performance from it through software

optimisation, which allowed us to greatly reduce the chipset cost.

How did the device gain from this strategy?

We were able to get from a lower-clocked processor the same performance as from higher-clocked ones. This reduced the power consumption, increasing the battery life. Moreover, it improved the charging circuit performance. For example, if you use the device for reading at night, you can get seven to eight hours of up-time.

Were there any difficulties in porting Android 2.2 onto an ARM9 processor without a GPU?

Getting an acceptable performance was a challenge. Initially, the touch response was not good enough. Then, the application took a long time to load, the graphics were slow and the video was jerky.

Why did you choose a resistive touchscreen instead of a capacitive one?

We designed this product to perform optimally for its intended use. It was not intended to compete with higher-end tablets but for use in enterprise environments where cost is important. We looked at where this tablet could be deployed meaningfully. As long as the use-cases are properly taken care, this device achieves its target.

One of the use-cases is conducting a survey. The device gives a good battery life as you take it around for survey. You have a questionnaire in front of you, with the answers